ICAD RT-5 · Set 1 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
5	a	b	c	d
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18	a	b	c	d
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Numerical — write the value

21	
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Chemistry

1	a	b	c	d
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3	a	b	c	d
4	a	b	c	d
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18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

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Mathematics

1	a	b	c	d
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3	a	b	c	d
4	a	b	c	d
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14	a	b	c	d
15	a	b	c	d
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17	a	b	c	d
18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

21	
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23	
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Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 1

Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

PART A · Q1–20 · SINGLE CORRECT

1. P2L1VSuggQ3 If $s = \sqrt[3]{t}$, then ds/dt is (a) (1/3)t^(2/3) (b) (1/3)t^(−2/3) (c) −(2/3)t^(−2/3) (d) (2/3)t^(−2/3)

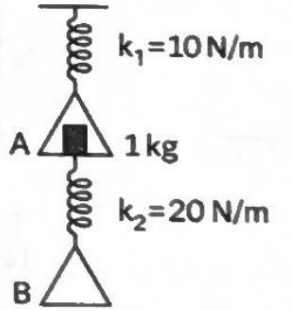
2. P4L3V3Q5 A stone falls freely from rest from a height h and it travels a distance 9h/25 in the last second. The value of h is: (take g = 9.8 m/s²) (a) 145 m (b) 100 m (c) 122.5 m (d) 200 m

3. P4L5V5Q5 The relation between distance x and time t is $t = \alpha x^2 + \beta x$ (α, β constants). Find the retardation of the particle: (a) $2\alpha v^3$ (b) $3\alpha v^3$ (c) $4\alpha v^3$ (d) $5\alpha v^3$

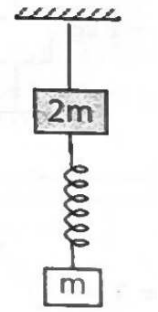
4. P4L10V5Q4 Two persons A and B stand at a distance x away from each other in an open field. Wind blows from A to B with a velocity w. (i) A beats a drum and sound travels from A to B in t_1 seconds. (ii) B beats a drum and sound travels from B to A in t_2 seconds. The wind velocity w is given by: (a) $2x/(t_1 + t_2)$ (b) $(x/2)[1/t_1 + 1/t_2]$ (c) $(x/2)[1/t_1 - 1/t_2]$ (d) $2x/\sqrt{(t_1 t_2)}$

5. P5L3V1Q4 Which of the following is NOT a condition for applying Lami's Theorem? (a) Forces must be non-collinear (b) Forces must be concurrent (c) Forces must be in equilibrium (d) Forces must be perpendicular

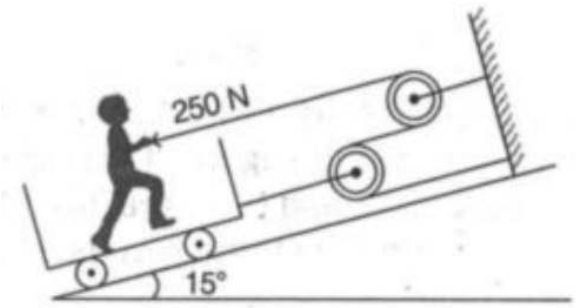
6. P6L5V6Q3 Two ideal springs with light pans are hanging vertically as shown: spring k_1 = 10 N/m from the ceiling holds pan A, and spring k_2 = 20 N/m hangs from pan A and holds pan B. 1 kg mass is placed in pan A; the descent of pan B in equilibrium is x_1. When the same mass is instead placed in pan B, the descent of pan B in equilibrium is x_2. Then x_2/x_1 is (a) 3/2 (b) 2/3 (c) 1/2 (d) None of these



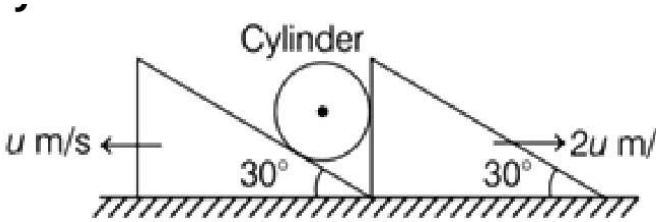
7. P6L6V4Q3 A block of mass 2m hangs from the ceiling by a string, and a block of mass m hangs from it by a light spring; the system is at rest. Initial acceleration of the lower block immediately after the string breaks: (a) 3g/2 upwards (b) 2g downwards (c) g/2 downwards (d) None of these



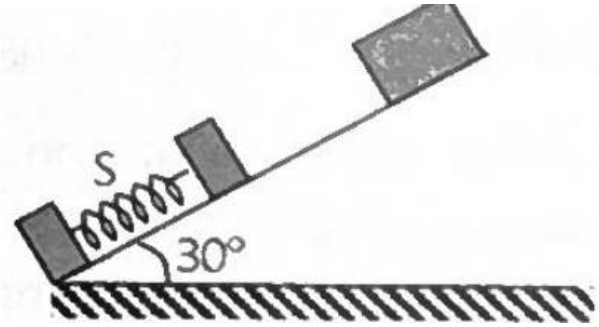
8. P6XVCAT4Q15 A trolley is being pulled up an inclined plane (inclination 15°) by a man sitting on it (as shown in figure). He applies a force of 250 N. If the combined mass of the man and trolley is 100 kg, the acceleration of the trolley will be [Use sin15° = 0.26, g = 10 m/s²] (a) 2.4 m/s² (b) 4 m/s² (c) 4.9 m/s² (d) 0.1 m/s²



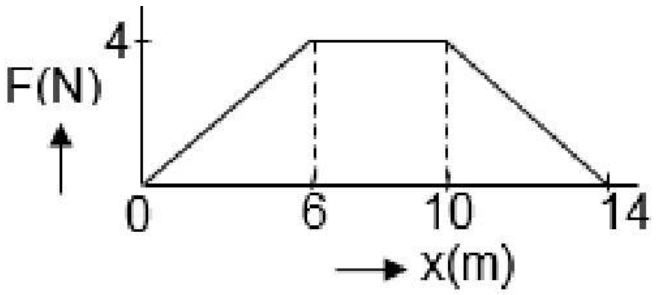
9. P6XVRT3Q11 System is shown in the figure. Assume that cylinder remains in contact with the two wedges. The velocity of cylinder is (a) $\sqrt{(19 - 4\sqrt{3})}\cdot u/2$ m/s (b) $\sqrt{13}\, u/2$ m/s (c) $\sqrt{3}$ u m/s (d) $\sqrt{7}$ u m/s



10. P8L6V5Q7 An ideal massless spring S can be compressed 1 m by a force of 100 N in equilibrium. The same spring is placed at the bottom of a frictionless inclined plane inclined at 30° to the horizontal. A 10 kg block M is released from rest at the top of the incline and is brought to rest momentarily after compressing the spring by 2 m. If g = 10 m/s², what is the speed of the mass just before it touches the spring? (a) √20 m/s (b) √30 m/s (c) √10 m/s (d) √40 m/s



11. P8XVCAT5Q5 The force acting on an object varies with the distance travelled by the object as shown in figure. Work done by the force in moving the object from x = 2 m to x = 4 m is : (a) 4 J (b) 6 J (c) 8 J (d) 2 J

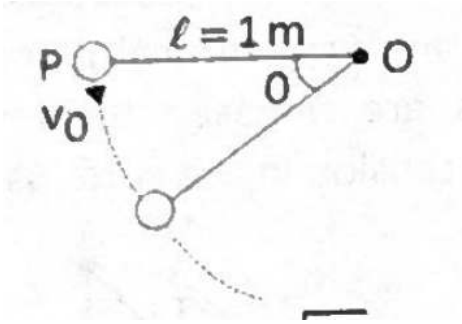


12. P9L2V5Q9 A body of mass 2 kg makes an elastic collision with a second body at rest and continues to move in the original direction but with one-fourth of its original speed. What is the mass of the second body? (a) 1.5 kg (b) 1.2 kg (c) 1.8 kg (d) 1.0 kg

13. P9L4V5Q7 Two billiard balls of mass 0.05 kg each moving in opposite directions with 10 ms -1 collide and rebound with the same speed. If the time duration of contact is t = 0.005s, then what is the force exerted on the ball due to each other? (a) 100 N (b) 200 N (c) 300 N (d) 400 N

14. P9SBVPYQQ5 An artillery piece of mass M 1 fires a shell of mass M 2 horizontally. Instantaneously after the firing, the ratio of kinetic energy of the artillery and that of the shell is: (a) M 1 /(M 1 +M 2) (b) M 2 /M 1 (c) M 2 /(M 1 +M 2) (d) M 1 /M 2

15. P10L5V5Q8 The sphere at P is given a downward velocity v 0 and swings in a vertical plane at the end of a rope of l = 1 m attached to a support at O. The rope breaks at angle 30° from horizontal, knowing that it can withstand a maximum tension equal to three times the weight of the sphere. Then the value of v 0 will be: (g = π² m/s²) (a) (g/2) m/s (b) (g/√2) m/s (c) √(3g/2) m/s (d) (g/3) m/s



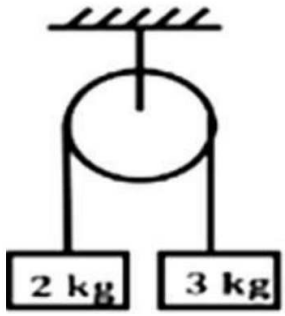
16. P10SBVPYQQ8 A car of 800 kg is taking a turn on a banked road of radius 300 m and angle of banking 30°. If the coefficient of static friction is 0.2 then the maximum speed with which the car can negotiate the turn safely: (g = 10 m/s², √3 = 1.73) (a) 264 m/s (b) 51.4 m/s (c) 70.4 m/s (d) 102.8 m/s

17. P10SBVPYQQ10 A small block of mass 100 g is tied to a spring of spring constant 7.5 N m −1 and length 20 cm. The other end of the spring is fixed at a particular point A. If the block moves in a circular path on a smooth horizontal surface with constant angular velocity 5 rad s −1 about point A, then the tension in the spring is: (a) 0.75 N (b) 0.25 N (c) 0.50 N (d) 1.5 N

18. P11L2VSuggQ2 A rod has mass 10 kg and length 4 m. What is its mass per unit length? (a) 0.4 kg/m (b) 2.5 kg/m (c) 4 kg/m (d) 10 kg/m

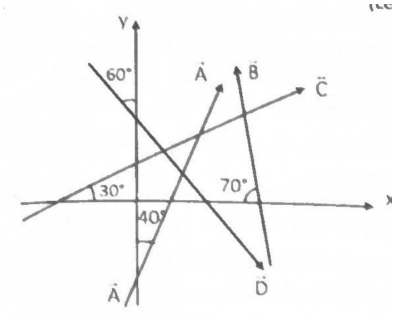
19. P11L3VSuggQ2 A body has two segments with areas A 1 = 4 m 2 , A 2 = 6 m 2 , and x-coordinates x 1 = 2 m, x 2 = 5 m. Find x COM . (a) 3.8 m (b) 4.2 m (c) 3.0 m (d) 5.0 m

20. P11XVCAT6Q11 Two blocks of masses 3 kg and 2 kg respectively are tied to the ends of a string which passes over a light pulley as shown in the figure below. The masses are held at rest at the same horizontal level and then released. The distance moved by COM in 5 seconds is (take g = 10 m/s²) (a) 10 m upward (b) 5 m downward (c) 10 m downward (d) 5 m upward



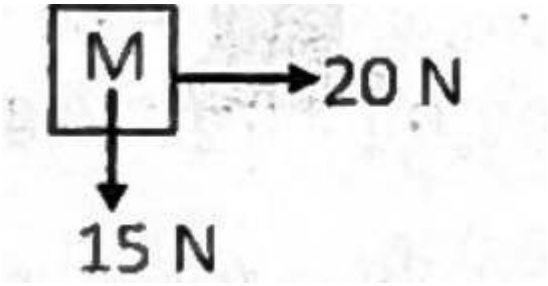
PART B · Q21–25 · NUMERICAL VALUE

21. P3L1VSuggQ5 Four vectors A, B, C, D are inclined with respect to the x-axis and y-axis as shown in the figure (A makes 40° with the y-axis, B makes 70° with the x-axis, C makes 30° with the x-axis, D makes 60° with the y-axis). Find the angle between vectors B and D, in degree.



22. P4XVCAT3Q19 A thief is running away on a straight road in a jeep moving with a speed of 9 ms^{-1}. A policeman chases him on a motor cycle moving at a speed of 10 ms^{-1}. If the instantaneous separation of the jeep from the motor cycle is 100 m, how long will it take for the policeman to catch the thief? (A) 1 s (B) 19 s (C) 90 s (D) 100 s

23. P7L2VSuggQ3 In the figure an object of mass M is kept on a rough table. Two forces are applied on it in the plane of the table: 20 N along one direction and 15 N perpendicular to it. The object does not move. Find the magnitude (in N) and direction of the static friction.



24. P7L3V5Q3 A block of mass 4 kg begins to slide (from rest) on a rough inclined plane and moves 1 metre in 0.707 seconds. Suppose the same block is instead pushed down the incline with a force of 4 N (parallel to the incline). How far will the block move in the first two seconds after starting from rest? (in metre)

25. P7XVRT4Q3 A block rests on a rough inclined plane making an angle of 30° with the horizontal. The coefficient of static friction between the block and the plane is 0.8. If the frictional force on the block is 10 N, the mass of the block (in kg) is (take g = 10 m/s 2) (A) 2.0 (B) 4.0 (C) 1.6 (D) 2.5

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L2VSuggQ2 At constant temperature, a sample of gas has a pressure of 1 atm and a volume of 4 L. What will be the pressure when the volume is decreased to 2 L? (a) 0.5 atm (b) 1 atm (c) 2 atm (d) 4 atm

2. C2L2V5Q4 What will be the energy of a photon corresponding to the wavelength 0.50 Å? (1 Å = 10^{-10} m, h = 6.626×10^{-34} J s, c = 3.0×10^{8} m/s) (a) 3.98×10^{-15} J (b) 3.00×10^{15} J (c) 3.90×10^{8} J (d) 3.00×10^{-34} J

3. C3L3V5Q3 The chloride of a metal contains 71% chlorine by mass and its vapour density is 50. The atomic mass of the metal is: (a) 29 (b) 58 (c) 35.5 (d) 71

4. C3SBVPYQQ10 The elemental composition of a compound is 54.2% C, 9.2% H and 36.6% O. If the molar mass of the compound is 132 g mol^{-1}, the molecular formula of the compound is: [Given: The relative atomic mass of C:H:O = 12:1:16] (a) $C_4H_9O_3$ (b) $C_6H_{12}O_6$ (c) $C_4H_8O_2$ (d) $C_6H_{12}O_3$

5. C4L5V4Q6 Which of the following elements F, P, S and Cl will have the most negative electron affinity and which the least negative? (a) F, Cl (b) Cl, F (c) Cl, S (d) Cl, P

6. C4L5V6Q7 The formation of the oxide ion $O^{2-}(g)$ requires first an exothermic and then an endothermic step: $O(g)+e^{-}\rightarrow O^{-}(g)$, $\Delta H = -142$ kJ mol^{-1}; $O^{-}(g)+e^{-}\rightarrow O^{2-}(g)$, $\Delta H = +748$ kJ mol^{-1}. This is because (a) O^{-} ion has comparatively larger size than the oxygen atom (b) oxygen has higher electron affinity (c) O^{-} ion will tend to resist the addition of another electron (d) oxygen is more electronegative

7. C4L5VSuggQ5 Consider $O(g) \rightarrow O^{-}(g)$ (ΔH_1) and $O^{-}(g) \rightarrow O^{2-}(g)$ (ΔH_2). Which statement is correct? (a) ΔH_1 and ΔH_2 are both negative (b) ΔH_1 is negative but ΔH_2 is positive due to electron–electron repulsion (c) ΔH_2 is negative because oxide ion is more stable (d) ΔH_2 is smaller in magnitude than ΔH_1

8. C4SBVPYQQ16 Consider the following elements In, Tl, Al, Pb, Sn and Ge. The most stable oxidation states of elements with highest and lowest first ionisation enthalpies, respectively, are: (a) +4 and +1 (b) +1 and +4 (c) +4 and +3 (d) +2 and +3

9. C5L1V3Q3 Which condition most strongly favours the formation of an ionic bond? (a) High IE and low lattice energy (b) Low IE, high EGE, high lattice energy (c) Low EGE and low lattice energy (d) High IE and high lattice energy

10. C5L2V3Q2 Find the formal charge on an oxygen atom in ozone (O_3) if it has 6 valence electrons, 4 non-bonding electrons, and is involved in 2 bonding pairs. (a) 0 (b) +1 (c) −1 (d) +2

11. C5L6V5Q4 In which of the following molecule, all the atoms lie in one plane? (a) CH_4 (b) BF_3 (c) PF_5 (d) NH_3

12. C5SBVPYQQ14 Among the following, the CORRECT combinations are: (A) IF_3 → T-shaped (sp^3d) (B) IF_5 → square pyramidal (sp^3d^2) (C) IF_7 → pentagonal bipyramidal (sp^3d^3) (D) ClO_4^- → square planar (sp^2d). Choose the correct answer: (a) A, B and C only (b) A and B only (c) A, B, C and D (d) B, C and D only

13. C6L1V6Q3 For a system in thermodynamic equilibrium, consider the following conditions: I. Temperature remains uniform throughout the system. II. No macroscopic property of the system changes with time. III. Mechanical equilibrium must be present, but chemical equilibrium is not necessary. IV. Pressure remains the same throughout the system. Which conditions must be satisfied? (a) I and II (b) I, II and IV (c) II and III (d) I, III and IV

14. C6L8V5Q3 If H 2 (g) + Cl 2 (g) → 2HCl; $\Delta H° = -44$ kcal and 2Na(s) + 2HCl(g) → 2NaCl(s) + H 2 (g); $\Delta H = -152$ kcal, then for Na(s) + 0.5Cl 2 (g) → NaCl(s); $\Delta H° = ?$ (a) 108 kcal (b) 196 kcal (c) −98 kcal (d) 54 kcal

15. C6L8V5Q9 Oxidising power of chlorine in aqueous solution can be determined by the parameters indicated below: (1/2)Cl 2 (g) → ((1/2)Δ diss H°) Cl(g) → (Δ eg H°) Cl − (g) → (Δ hyd H°) Cl − (aq). The energy involved in the conversion of (1/2)Cl 2 (g) to Cl − (aq) (using the data Δ diss H° Cl 2 = 240 kJ mol −1 , Δ eg H° Cl = −349 kJ mol −1 , Δ hyd H° Cl − = −381 kJ mol −1) will be: (a) +120 kJ mol −1 (b) +152 kJ mol −1 (c) −610 kJ mol −1 (d) −850 kJ mol −1

16. C6L9VSuggQ1 The heat change for the reaction H 2 O(l) → H 2 O(g) is called as: (a) heat of vaporisation (b) heat of solution (c) heat of fusion (d) heat of formation

17. C6L10V3Q2 For a natural irreversible process, the entropy change of the universe is: (a) Zero (b) Positive (c) Negative (d) Constant

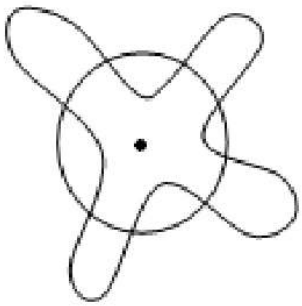
18. C6L11VSuggQ3 For a spontaneous reaction ΔG, equilibrium constant (K) and E° cell will be respectively: (a) −ve, >1, +ve (b) +ve, >1, −ve (c) +ve, <1, −ve (d) −ve, >1, −ve

19. C6L12V4Q6 All the naturally occurring processes, i.e. spontaneous, proceed spontaneously in a direction which leads to: (a) decrease of free energy (b) increase of free energy (c) decease of entropy (d) increase of enthalpy

PART B · Q21–25 · NUMERICAL VALUE

21. C1XVRT2Q39 Between two horizontal plates separated by 2.0 mm, a liquid of viscosity 3.0 poise fills the gap. The upper plate (area 40.0 cm^2) moves at 6.0 cm s^{-1} relative to the lower plate. Calculate the viscous force on the upper plate. (Use the cgs formula $F = \eta \times (A \cdot v / x)$) (A) 1,800 dyn (B) 3,600 dyn (C) 7,200 dyn (D) 360 dyn

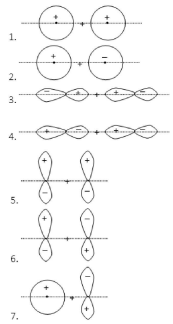
22. C2XVCAT3Q46 In H-atom, the motion of an electron is represented as shown in the figure (a de Broglie standing wave). The number of revolutions/sec made by such electron is equal to $x \times 10^{14}$. The value of x is (round off to nearest integer)



20. C6XVCAT6Q39 Heat of combustion of CH_4, C_2H_6, C_2H_4 and C_2H_2 are −212.8, −373.0, −337.0 and −310.5 kcal mol^{-1} respectively at the same temperature. The best fuel among these gases is (Atomic masses: H = 1, C = 12) (a) CH_4 (b) C_2H_6 (c) C_2H_4 (d) C_2H_2

23. C3XVRT2Q49 Each copper (II) sulphate unit is associated with water molecules. When this compound is heated in air above 100°C, it loses the water molecules and also its colour: $CuSO_4 \cdot xH_2O \rightarrow CuSO_4 + xH_2O$. If 9.60 g of $CuSO_4$ are left after heating 15.01 g of the blue compound, calculate the number of moles of H_2O originally present per mole of $CuSO_4$ in the compound (i.e. x).

24. C5XVCAT5Q50 How many of the following combination will form σ bond.



25. C6XVCAT6Q29 Enthalpy of neutralization of H 3 PO 3 (dibasic acid) is −105 kJ/mol using KOH. Enthalpy of neutralization of HCl by KOH is −56 kJ/mol. Calculate ΔH ionization of H 3 PO 3 into its ions: (A) 7 kJ/mol (B) 5 kJ/mol (C) 2.5 kJ/mol (D) 3.5 kJ/mol

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L1V3Q4 The angle subtended at the centre of a circle of diameter 50 cm by an arc of 11 cm is (take π = 22/7) (a) 25°10' (b) 20°12' (c) 25°12' (d) 20°10'

2. M2SBVPYQQ4 Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of B. Then the distance of the point P(m, n) from the point Q(−2, −3) is (a) 10 (b) 6 (c) 4 (d) 8

3. M4L1V5Q8 If 5^((1−x)/3) = 25^(6(x+1)), then the value of x is (a) 37/35 (b) −35/37 (c) 35 (d) 5

4. M4XVRT4Q54 If $f(x) = 4^x - 2^x + 1$ then value of f(−2) is (a) 16/13 (b) 13/16 (c) 12/23 (d) 1

5. M5AL3VSuggQ5 The minimum value of the expression $(9x^2 \sin^2 x + 4)/(x \sin x)$, $x \in (0, \pi)$ is: (a) 16/3 (b) 6 (c) 12 (d) 8/3

6. M5BL1V5Q9 If $\sin x + \sin y = \sqrt{3}(\cos y - \cos x)$, then $\sin 3x + \sin 3y$ = (Note: $x - y \in (0, \pi)$) (a) 2 sin 3x (b) 0 (c) 1 (d) none

7. M5CL1VSuggQ4 If A lies in the second quadrant and sinA = 5/13, determine the sign of cos 2A. (a) Positive (b) Negative (c) Zero (d) Cannot be determined

8. M6L5VSuggQ2 Which of the following is a rational inequality? (a) $x^2 - 4 < 0$ (b) $(x - 1)/(x + 2) > 0$ (c) $x + 5 > 0$ (d) $x^3 \le 0$

9. M7L1V5Q7 The product of all real roots of (25/16)^(x²−2x−3) = (4/5)^(x+1) is: (a) 5/2 (b) −5/2 (c) 2 (d) −2

10. M7SBVPYQQ1 The value of ∑_{k=1}^∞ (−1)^{k+1}·(k(k+1)/k!) is. (a) e/2 (b) 2/e (c) √e (d) 1/e

11. M7XVCAT4Q69 If x_1, x_2 be the solutions of the equation (x + 1)^(log10(x + 1)) = 100(x + 1) then value of $x_1 + x_2$ is (a) 0.981 (b) 9.81 (c) 98.1 (d) 981

12. M8ASBVPYQQ10 Suppose a_1, a_2, ..., a n , ... be an arithmetic progression of natural numbers. If the ratio of the sum of the first five terms to the sum of first nine terms of the progression is 5 : 17 and $110 < a_{15} < 120$, then the sum of the first ten terms of the progression is equal to (a) 290 (b) 380 (c) 460 (d) 510

13. M8AXVRT4Q68 The common difference of the A.P. b_1, b_2, ..., b_m is 2 more than the common difference of A.P. a_1, a_2, ..., a_n. If $a_{40} = -159$, $a_{100} = -399$ and $b_{100} = a_{70}$, then b_1 is equal to (a) 127 (b) 81 (c) −127 (d) −81

14. M8BL1V3Q2 If the sequence 2, 4, 8, 16, ... contains 15 terms then the 8 th term from last is (a) 1024 (b) 128 (c) 256 (d) 512

15. M8CL1V3Q5 If x + y = a and $x^2 + y^2 = b$, then the value of $(x^3 + y^3)$ is (a) ab (b) $a^2 + b^2$ (c) $a + b^2$ (d) $(3ab - a^3)/2$

16. M8DL1V2Q3 Find the sum of cubes of first 2 natural numbers. (a) 9 (b) 8 (c) 1 (d) 3

17. M8DL2V2Q3 Find 1 st term of A.G.P. for A.P. = 1, 2,... and G.P. = 2, 4,... (a) 1 (b) 2 (c) 4 (d) 3

18. M9L1V3Q4 i + 1/i = (a) 1 (b) -1 (c) 0 (d) 2i

19. M9L1V5Q5 The least positive integer 'n' such that ((1-i)/(1+i))^(2n) = 1 is (a) 2 (b) 4 (c) 6 (d) 14

20. M9L4V1Q3 The triangle inequality applies to: (a) Real numbers only (b) Imaginary numbers only (c) Complex numbers (d) Rational numbers only

PART B · Q21–25 · NUMERICAL VALUE

21. M3XVCAT3Q68 If $\sin x + \sin^2 x = 1$, then $\cos^8 x + 2\cos^6 x + \cos^4 x =$ (A) 2 (B) 1 (C) 3 (D) 1/2

22. M5AXVRT4Q64 If $\tan 48^\circ \tan 23^\circ \tan 42^\circ \tan 67^\circ = \tan(A + 30^\circ)$, then A will be (A) 30° (B) 15° (C) 45° (D) 60°

23. M6XVRT4Q63 The number of integral solutions for $||x - 2015| - 2015| \le 1000$ is (A) 4000 (B) 4001 (C) 4002 (D) 4003

24. M8DL5VSuggQ9 Find the minimum value of the sum of real numbers a^(−5), a^(−4), 3a^(−3), 1, a^(8) and a^(10) with a > 0.

25. M9L5V5Q7 If the real part of the complex number $z = (3 + 2i\cos\theta)/(1 - 3i\cos\theta)$, $\theta \in (0, \pi/2)$ is zero, then the value of $\sin^2 3\theta + \cos^2\theta$ is equal to _____